

## MARINGA, Brazil

WMO: 868990

Lat: 23.405S

Lon: 51.933W

Elev: 544

StdP: 94.96

Time Zone: -3.00 (SAB)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 6             | 8.0        | 10.0 | 1.8                           | 4.6 | 15.5 | 3.9 | 5.3 | 17.5 | 9.0                   | 21.6 | 8.3 | 20.7 | 3.1                   | 210  | 0.353 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.7                    | 35.0            | 21.2 | 33.9 | 21.5 | 32.9 | 21.6 | 25.5                | 29.6 | 24.9 | 29.2 | 24.4 | 28.7 | 4.8                  | 340  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.2                            | 20.5 | 26.7 | 24.0 | 20.1 | 26.4 | 23.2 | 19.2 | 26.1 | 81.8          | 29.8 | 79.4 | 29.2 | 77.3 | 28.7 | 28.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.1           | 8.1  | 7.2 | 4.9                        | 37.0 | 1.4                | 1.0 | 3.9                                                | 37.7 | 3.1        | 38.2 | 2.3        | 38.8 | 1.4        | 39.5 |    |    |
| (5)               |      |     | 4.0                        | 26.5 | 1.8                | 0.8 | 2.7                                                | 27.1 | 1.7        | 27.5 | 0.7        | 28.0 | -0.6       | 28.5 |    |    |

## Monthly Climatic Design Conditions

|             |                                                                                  | Annual  | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|-------------|----------------------------------------------------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|             |                                                                                  | (d)     | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)         | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg   | 23.1  | 25.5  | 26.0  | 25.2  | 23.3  | 20.3  | 19.0  | 19.6  | 21.0  | 23.2  | 24.0  | 24.9  | 25.3 | (6)  |
| (7)         |                                                                                  | DBStd   | 3.80  | 1.77  | 2.02  | 1.96  | 2.25  | 2.99  | 3.23  | 3.74  | 4.28  | 4.17  | 3.15  | 2.33  | 2.00 | (7)  |
| (8)         |                                                                                  | HDD10.0 | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)         |                                                                                  | HDD18.3 | 114   | 0     | 0     | 0     | 1     | 16    | 31    | 33    | 25    | 7     | 1     | 0     | 0    | (9)  |
| (10)        |                                                                                  | CDD10.0 | 4782  | 479   | 449   | 470   | 400   | 321   | 270   | 299   | 341   | 396   | 436   | 447   | 474  | (10) |
| (11)        |                                                                                  | CDD18.3 | 1854  | 221   | 216   | 212   | 151   | 79    | 51    | 72    | 107   | 153   | 178   | 197   | 217  | (11) |
| (12)        |                                                                                  | CDH23.3 | 15027 | 1704  | 1799  | 1698  | 1048  | 405   | 237   | 438   | 936   | 1567  | 1664  | 1742  | 1788 | (12) |
| (13)        | CDH26.7                                                                          | 5537    | 604   | 735   | 629   | 311   | 68    | 29    | 80    | 339   | 693   | 717   | 688   | 644   | (13) |      |
| (14)        | Wind                                                                             | WSAvg   | 3.4   | 3.2   | 3.3   | 3.2   | 3.0   | 3.0   | 3.1   | 3.3   | 3.8   | 3.9   | 3.7   | 3.6   | 3.1  | (14) |
| (15)        | Precipitation                                                                    | PrecAvg | 1631  | 229   | 186   | 138   | 102   | 137   | 98    | 78    | 58    | 134   | 145   | 149   | 177  | (15) |
| (16)        |                                                                                  | PrecMax | 2242  | 379   | 324   | 276   | 270   | 349   | 338   | 267   | 193   | 325   | 330   | 405   | 287  | (16) |
| (17)        |                                                                                  | PrecMin | 1272  | 63    | 43    | 69    | 8     | 34    | 6     | 9     | 1     | 15    | 42    | 45    | 31   | (17) |
| (18)        |                                                                                  | PrecStd | 243   | 82    | 75    | 55    | 53    | 85    | 77    | 64    | 53    | 72    | 80    | 73    | 69   | (18) |
| (19)        | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%    | DB    | 34.0  | 36.0  | 34.0  | 32.1  | 30.2  | 28.8  | 30.0  | 33.8  | 36.2  | 36.2  | 34.9  | 34.1 | (19) |
| MCWB        |                                                                                  |         | 23.2  | 21.4  | 22.2  | 22.1  | 21.3  | 19.4  | 18.2  | 18.5  | 19.9  | 21.5  | 23.0  | 21.9  | (20) |      |
| 2%          |                                                                                  | DB      | 32.3  | 34.0  | 32.9  | 30.9  | 28.2  | 27.2  | 28.8  | 31.9  | 34.5  | 34.8  | 33.2  | 32.8  | (21) |      |
|             |                                                                                  | MCWB    | 23.5  | 22.8  | 22.3  | 21.9  | 20.8  | 19.1  | 18.0  | 18.2  | 19.8  | 21.4  | 22.1  | 22.9  | (22) |      |
| 5%          |                                                                                  | DB      | 31.1  | 32.8  | 31.6  | 29.9  | 27.1  | 26.1  | 27.2  | 30.2  | 32.9  | 32.8  | 31.9  | 31.2  | (23) |      |
|             |                                                                                  | MCWB    | 23.5  | 23.1  | 22.4  | 21.3  | 20.0  | 18.8  | 17.5  | 17.9  | 19.0  | 20.8  | 21.2  | 22.9  | (24) |      |
| 10%         |                                                                                  | DB      | 30.0  | 31.1  | 30.2  | 28.8  | 25.9  | 24.3  | 26.0  | 28.9  | 30.9  | 30.8  | 30.6  | 30.1  | (25) |      |
|             |                                                                                  | MCWB    | 23.5  | 23.1  | 22.2  | 20.9  | 19.5  | 18.6  | 17.4  | 17.7  | 18.8  | 20.7  | 21.1  | 22.6  | (26) |      |
| (27)        | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%    | WB    | 26.2  | 25.6  | 25.4  | 24.4  | 23.2  | 22.8  | 21.2  | 21.6  | 23.5  | 24.4  | 25.2  | 26.4 | (27) |
| MCDWB       |                                                                                  |         | 30.0  | 30.4  | 29.4  | 27.8  | 27.2  | 25.2  | 24.8  | 26.8  | 31.2  | 29.5  | 30.7  | 29.5  | (28) |      |
| 2%          |                                                                                  | WB      | 25.4  | 25.0  | 24.6  | 23.7  | 22.4  | 21.4  | 20.1  | 19.8  | 22.1  | 23.4  | 24.4  | 25.2  | (29) |      |
|             |                                                                                  | MCDWB   | 29.3  | 29.9  | 29.0  | 27.7  | 26.3  | 23.8  | 24.4  | 27.2  | 28.9  | 29.8  | 29.5  | 29.1  | (30) |      |
| 5%          |                                                                                  | WB      | 24.8  | 24.5  | 23.9  | 23.0  | 21.5  | 20.2  | 19.1  | 19.0  | 21.1  | 22.6  | 23.7  | 24.6  | (31) |      |
|             |                                                                                  | MCDWB   | 28.7  | 29.0  | 28.5  | 27.0  | 25.0  | 23.4  | 23.7  | 26.4  | 27.3  | 29.0  | 28.4  | 28.5  | (32) |      |
| 10%         |                                                                                  | WB      | 24.2  | 24.0  | 23.2  | 22.2  | 20.6  | 19.4  | 18.4  | 18.2  | 20.1  | 21.8  | 22.9  | 23.9  | (33) |      |
|             |                                                                                  | MCDWB   | 28.1  | 28.2  | 27.7  | 25.9  | 23.9  | 22.7  | 23.3  | 25.8  | 26.2  | 27.6  | 27.3  | 27.6  | (34) |      |
| (35)        | Mean Daily<br>Temperature<br>Range                                               | 5% DB   | MDBR  | 9.1   | 9.7   | 9.8   | 9.9   | 9.1   | 9.1   | 10.4  | 11.4  | 11.9  | 10.9  | 10.6  | 9.3  | (35) |
| MCDBR       |                                                                                  |         | 10.7  | 11.8  | 11.4  | 11.0  | 10.2  | 10.3  | 11.4  | 12.5  | 12.9  | 13.3  | 12.3  | 11.0  | (36) |      |
| MCWBR       |                                                                                  |         | 4.7   | 4.0   | 4.0   | 4.7   | 4.7   | 4.8   | 4.7   | 4.9   | 4.9   | 5.2   | 4.5   | 4.9   | (37) |      |
| 5% WB       |                                                                                  | MCDBR   | 9.1   | 9.5   | 9.5   | 9.3   | 8.8   | 8.8   | 10.1  | 11.5  | 11.3  | 11.0  | 9.9   | 9.1   | (38) |      |
|             |                                                                                  | MCWBR   | 4.6   | 4.1   | 4.1   | 4.6   | 4.7   | 4.9   | 5.0   | 5.3   | 5.2   | 4.9   | 4.5   | 4.8   | (39) |      |
| (40)        | Clear-Sky<br>Solar<br>Irradiance                                                 | taub    | 0.395 | 0.391 | 0.388 | 0.368 | 0.351 | 0.368 | 0.368 | 0.428 | 0.532 | 0.455 | 0.405 | 0.399 | (40) |      |
| taud        |                                                                                  | 2.425   | 2.434 | 2.434 | 2.477 | 2.483 | 2.433 | 2.416 | 2.201 | 1.921 | 2.211 | 2.356 | 2.395 | (41)  |      |      |
| Ebn at Noon |                                                                                  | 950     | 939   | 915   | 890   | 864   | 824   | 837   | 816   | 770   | 870   | 933   | 947   | (42)  |      |      |
| Edh at Noon |                                                                                  | 125     | 121   | 116   | 102   | 93    | 94    | 99    | 133   | 190   | 150   | 133   | 129   | (43)  |      |      |
| (44)        | All-Sky Solar<br>Radiation                                                       | RadAvg  | 5.92  | 5.78  | 5.46  | 4.79  | 3.73  | 3.40  | 3.67  | 4.48  | 4.96  | 5.42  | 6.06  | 6.17  | (44) |      |
| (45)        |                                                                                  | RadStd  | 0.56  | 0.47  | 0.38  | 0.33  | 0.41  | 0.35  | 0.39  | 0.31  | 0.49  | 0.45  | 0.50  | 0.60  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page